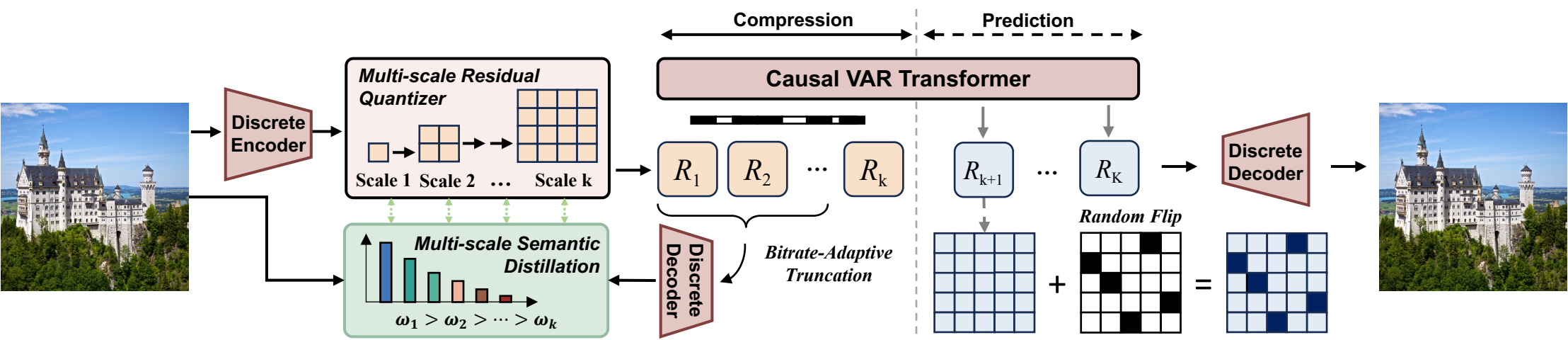

Compression
Prediction
Discrete Encoder
Multi-scale Residual Quantizer
Scale 1
Scale 2
…
Scale k
Causal VAR Transformer
R_1
R_2
…
R_k
R_{k+1}
…
R_K
Discrete Decoder
Discrete Decoder
Multi-scale Semantic Distillation
$\omega_1 > \omega_2 > \cdots > \omega_k$
Bitrate-Adaptive Truncation
Random Flip
+
=